

Tech & Teach

Digital Chip Design Program

RTL Design

Task #12

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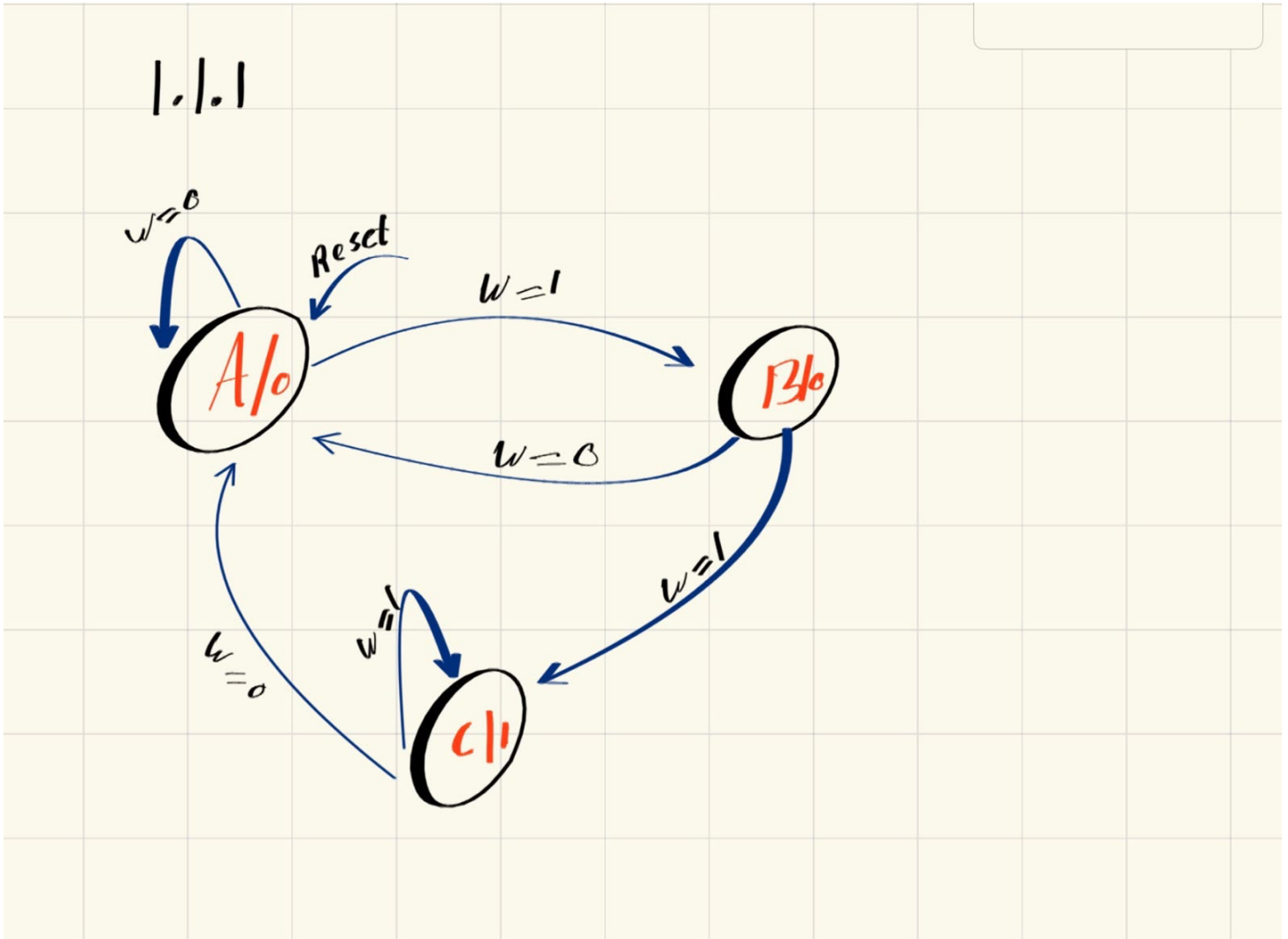
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Controller – Part 1

Version 1.0

Task #1.1

1.1.1



1.1.2

Moore FSM

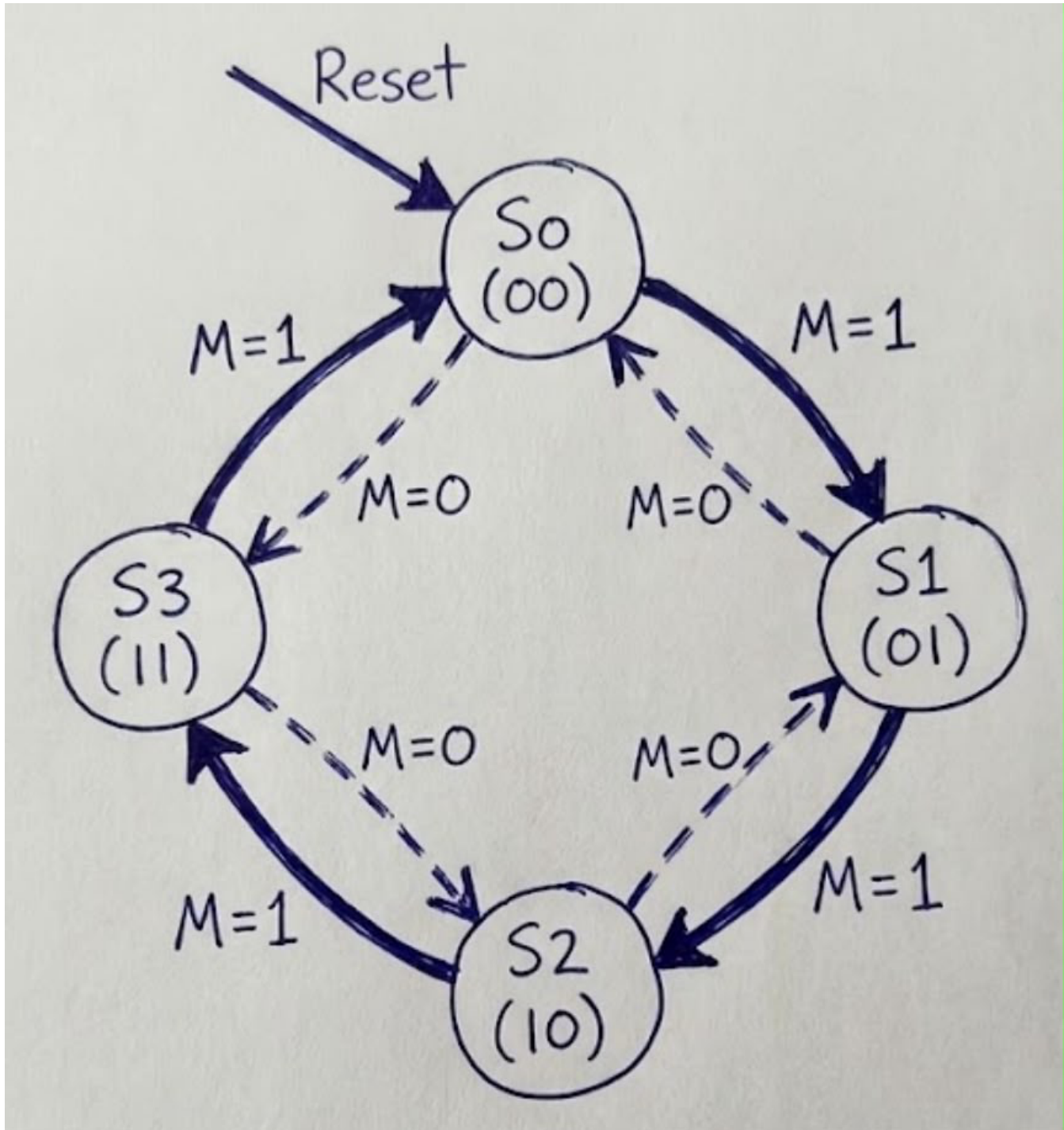
Task #1.2

```
1 module FSM_Mealy(  
2     input logic clk,rstn , in,  
3     output logic out  
4 );  
5  
6     enum logic [1:0]{  
7         A ,  
8         B ,  
9         C ,  
10        unused = 2'bx  
11    } current_state, next_state;  
12  
13    always_comb begin  
14        case (current_state)  
15            A : begin  
16                next_state = in ? C : B;  
17                out = 1'b0;  
18            end  
19  
20            B : begin  
21                next_state = in ? C : B;  
22                out = in ? 1'b0 : 1'b1;  
23            end  
24  
25            C : begin  
26                next_state = in ? C : B;  
27                out = in ? 1'b1 : 1'b0;  
28            end  
29  
30            default: begin  
31                next_state = unused;  
32                out = 1'bx;  
33            end  
34        endcase  
35    end  
37    always @(posedge clk or negedge rstn )  
38        if(!rstn)  
39            current_state <= A;  
40        else  
41            current_state <= next_state;  
42    endmodule
```



Task #1.3

1.3.1



1.3.2

```
1 module fsm(  
2     input logic clk,  
3     input logic rstn,  
4     input logic M,  
5     output logic [1:0] Q  
6 );  
7  
8 // states  
9     enum logic [1:0]{  
10         S0 = 2'b00,  
11         S1 = 2'b01,  
12         S2 = 2'b10,  
13         S3 = 2'b11  
14     } current_state, next_state;  
15  
16 // output logic  
17     always_comb begin  
18         if(current_state==S0)  
19             Q = 2'b00;  
20         else if(current_state==S1)  
21             Q = 2'b01;  
22         else if(current_state==S2)  
23             Q = 2'b10;  
24         else  
25             Q = 2'b11;  
26     end  
27  
28 // Next state logic  
29     always_comb begin  
30         if(current_state==S0) begin  
31             if(M)  
32                 next_state = S1;  
33             else  
34                 next_state = S3;  
35         end  
36
```

```

37  else if(current_state==S1) begin
38      if(M)
39          next_state = S2;
40      else
41          next_state = S0;
42      end
43
44  else if(current_state==S2) begin
45
46      if(M)
47          next_state = S3;
48      else
49          next_state = S1;
50      end
51
52  else begin
53      if(M)
54          next_state = S0;
55      else
56          next_state = S2;
57      end
58  end
59
60  //State register
61  always_ff @(posedge clk or negedge rstn) begin
62      if(!rstn)
63          current_state <= S0;
64      else
65          current_state <= next_state;
66      end
67  endmodule

```

Optional task #XX

```
1 module FSM_Moore(
2     input logic clk,rstn , in,
3     output logic out
4 );
5
6     enum logic [2:0]{
7         A ,
8         B ,
9         C ,
10        D ,
11        E ,
12        unused = 3'bxx
13    } current_state, next_state;
14
15    always_comb begin
16        case (current_state)
17            A : out = 0;
18            B : out = 0;
19            C : out = 1;
20            D : out = 0;
21            E : out = 1;
22            default: out = 1'bxx;
23        endcase
24    end
25
26    always_comb begin
27        case (current_state )
28            A : next_state = in ? D : B;
29            B : next_state = in ? D : C;
30            C : next_state = in ? D : C;
31            D : next_state = in ? E : B;
32            E : next_state = in ? E : B;
33            default: next_state = unused;
34        endcase
35    end
36
37    always_ff @(posedge clk or negedge rstn ) begin
38        if(!rstn)
39            current_state <= A;
40        else
41            current_state <= next_state;
42    end
43 endmodule
```

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Work distribution

Section	Farah	Joud	Tala	Horiah	Rana	Hoor	Sandra
Report	✓	✓	✓	✓	✓	✓	✓
Task 1.1			✓	✓			
Task 1.2					✓	✓	✓
Task 1.3	✓	✓					
Optional task						✓	